Lab 1 Report

Resistive Respiration Sensor

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1 Sensor Test

1.1 Voltage Divider & Resistance - Output Relation

The Arduino Output is 10-Bit, thus the maximum ADC value is given by

$$ADC_{max} = 2^{10} - 1 = 1023$$

The formula for the voltage divider can be rewritten in terms of the ADC value, where the V_{out} is replaced by ADC_{out} and V_{in} by ADC_{max} with $R_1 = 10\text{k}\Omega$ representing the fixed resistor:

$$ADC_{out} = \frac{R_{sensor}}{R_1 + R_{sensor}} ADC_{max}$$

$$R_{sensor} = R_1 \frac{ADC_{out}}{ADC_{max} - ADC_{out}}$$

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1.2 20 % Strain - Pre-Conditioning

Figure 1 shows the sensor output when being stretched to 20% and relaxed to 0% strain four times in a row. The sensors resistance rises significantly, as expected, when under strain. Note worthy is also that the initial resistance was much lower then it was at the intermediate 0% strain sections. This can be explained by conductive paths in the sensor that are only reforming slowly when the sensor is at rest.

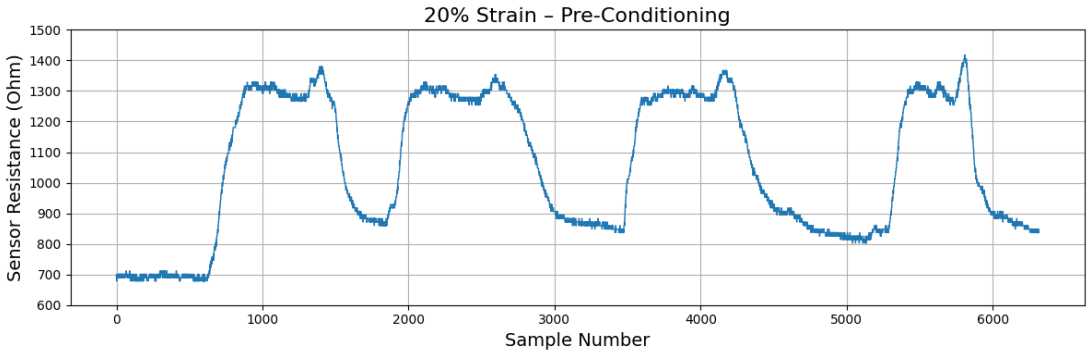


Figure 1: Unconditioned sensor output under 20% strain

1.3 Conditioning & 20 % Strain - Post-Conditioning

The sensor was conditioned by stretching it multiple times to 50% strain as show in Figure 2. Repeating the 20% strain test afterwards did not show large differ-

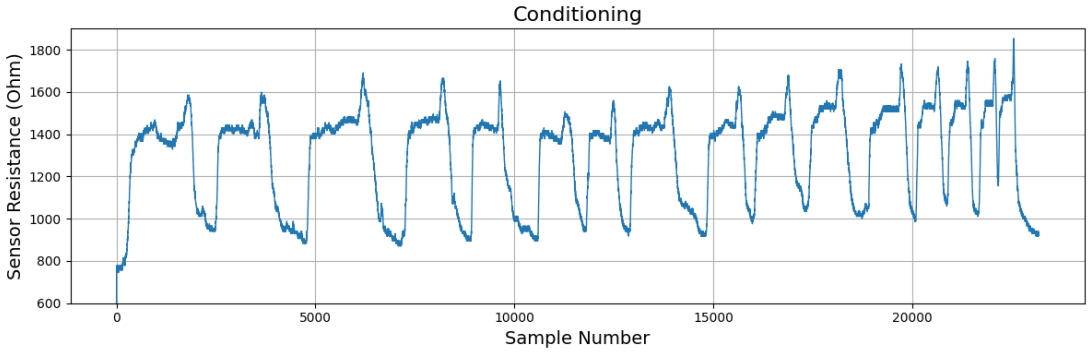


Figure 2: Conditioning the sensor by repeated 50% strain

ences. The only change that can be observed in Figure 3 is that the resistance at the intermediate 0% strain segments has further increase to above 1 $k\Omega$ compared to the preconditioned state $\approx 900\,\Omega$

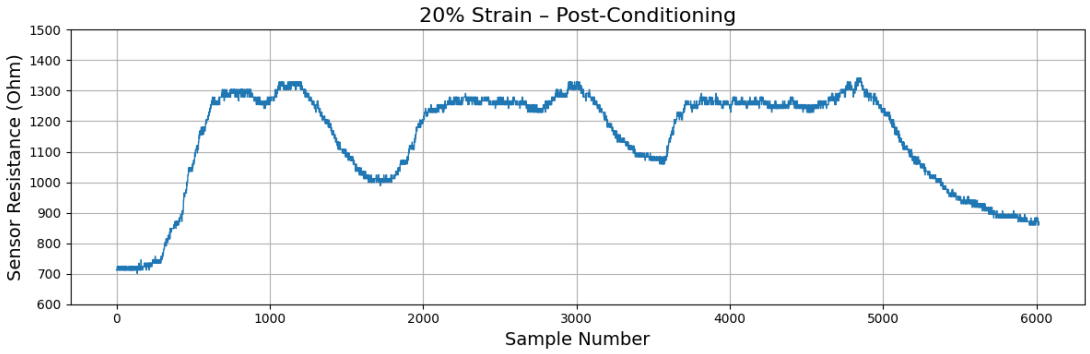


Figure 3: Conditioned sensor output under 20% Strain

1.4 Strain-dependent Sensor Values & Calibration Curve

The plot in Figure 4 shows the raw sensor data of five repetitions of the test from task 4. The sensor was stretched to 10%, 20% and 30% strain with a holding time of five seconds at each strain level before it was relaxed the same way. Due to the signal noise we applied an moving average filter with a window size of 100 samples. The smoothed sensor data are shown in Figure 5

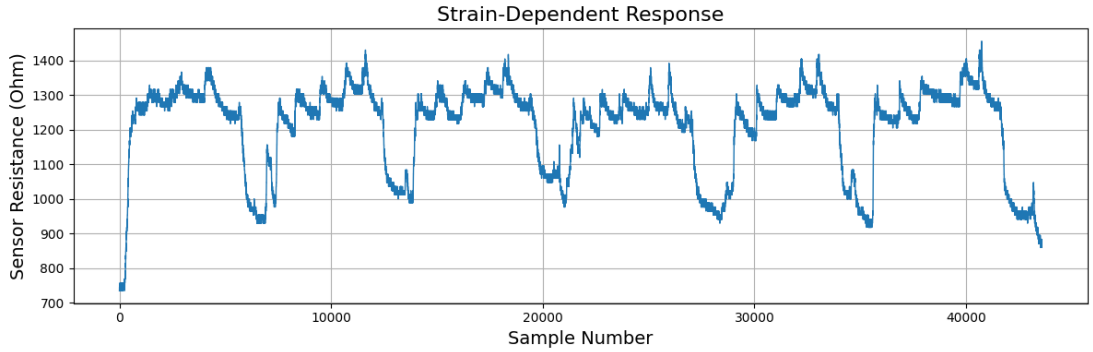


Figure 4: Sensor output when stretched to and relaxed from 30% strain in steps of 10% with a 5 s hold at each step - raw sensor data

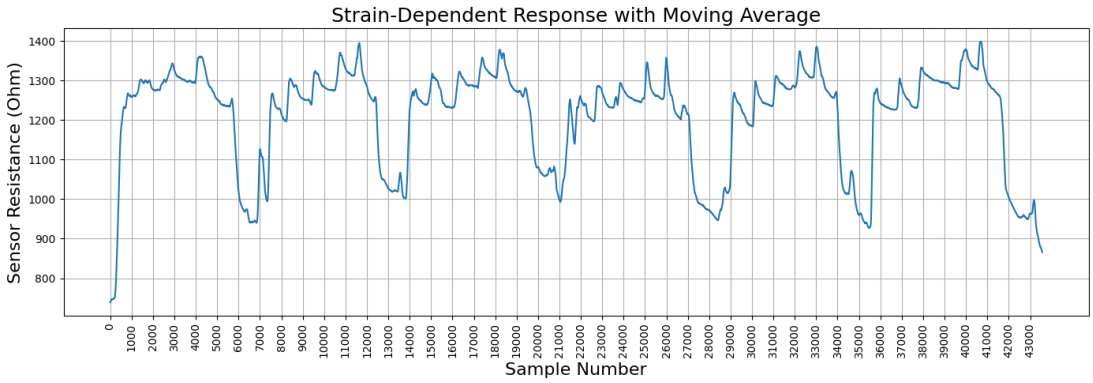


Figure 5: sensor output when stretched to and relaxed from 30% strain in steps of 10% with a 5 s hold at each step - filtered data

The calculations of the standard deviation and the mean resistance was performed on the repetitions three, five and six and the sample intervals for the corresponding strain levels have been selected manually.

The Table 1 shows the mean of each of the strain levels that was calculated on the combined sensor outputs of all of the three selected repetitions of the test. Therefore, and due to the previous filtering the STDW now measures the inter-circle variance and is a measure of how reproducible the sensor works. The used formulas are:

$$\text{Mean: } \bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\text{STDW: } \sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

Noteworthy, is that the plot Figure 5 and the values in Table 1 show that the

	0 % ext.	10 % ext.	20 % ext.	30 %	20 % rec.	10 % rec.	0 % rec.
Mean [Ω]	985.42	1239.66	1259.55	1295.23	1334.12	1287.17	1013.05
STDW [Ω]	37.48	24.30	27.08	13.67	23.35	43.63	51.89

Table 1: Mean resistance and standard deviation at different strain levels during extension and recovery.

resistance values rise to an initial peak and then decrease while the sensor is hold at a constant strain resulting in a large STDW. Furthermore, the resistance values during the release phases of the test still show higher resistance values then for the maximum strain of 30 %. This might be explained by a hysteresis of the sensor resistance due to slow recovering conductive connections or by a baseline shift of the minimal resistance due to a slipping of the clamps on the sensor under the continuous strain.

From the mean resistance value for each strain level the the calibration curve can be plotted and is shown in Figure 6. The calibration curve is non-linear with

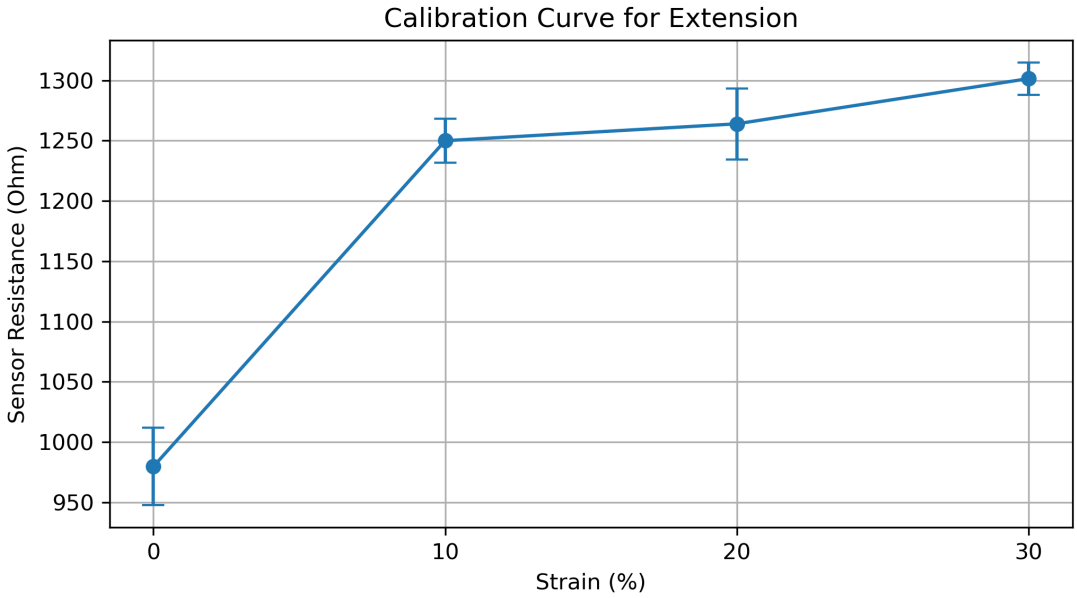


Figure 6: sensor calibration curve

much lower sensibility for higher strains, thus the initial tightness of the sensor during the wearable data collection should be chosen carefully to ensure reliable.

2 Wearable Data Collection

2.1 Variation of Sensor Position & Tightness

Different tests were conducted while varying the position on/under the clothing and also while varying the level of tightness. The breath depth was changed within the tests themselves, maintaining deep breath in the first half and normal breaths in the second half. More specifically:

- On the clothing, low tightness
- Under the clothing, low tightness
- On the clothing, high tightness

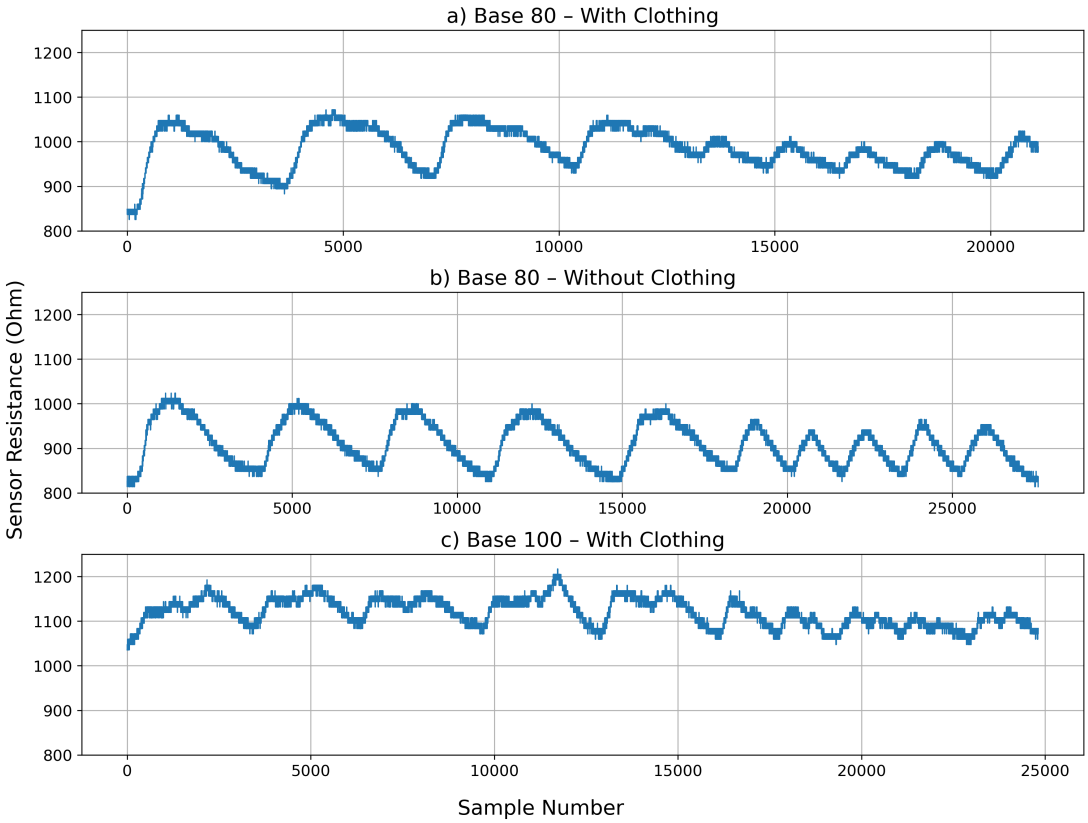


Figure 7: Raw data of the wearable sensor in varying conditions, a) on the clothing, low tightness, b) under the clothing, low tightness c) on the clothing, high tightness

Plots a) and b) in Figure 7 highlight the fact that at a low level of tightness, independently of shallow or deep breathing, it is possible to accurately monitor breathing. Additionally, being worn on top or under the clothing doesn't seem to play a role, as long as the initial tightness level is the same.

Conversely, when the tightness level is higher(third plot), the monitor of the breathing becomes harder(even though not impossible) as the variations in the measured resistance become in relative terms smaller. Indeed, this is additionally confirmed when paying attention to the difference between shallow and deep breaths; shallow breaths, which cause a smaller change in Resistance, are in fact harder to identify then deep breaths, which instead consist in a bigger change in the Resistance value. These results align with the lower sensibility already seen in the calibration curve (Figure 6).

2.2 Raw Data Comparison of Breathing Depth

In Figure 8 one can observe the raw signals of 2 different tests conducted, respectively, with Deep breathing and Shallow breathing. Already from a naked-eye

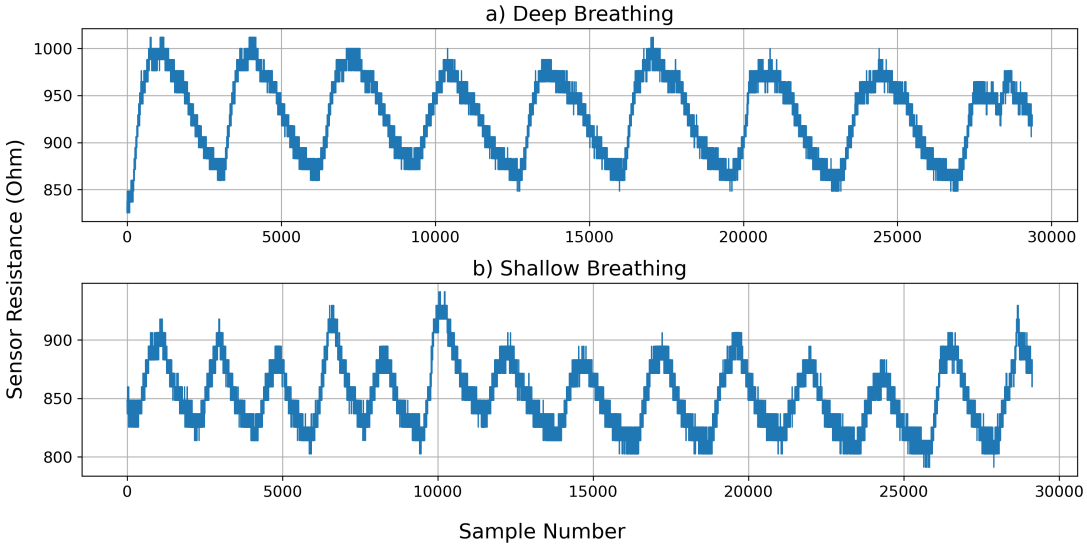


Figure 8: Raw data of the wearable sensor for varying breathing depth

analysis, one can observe that a series of peaks, corresponding to the inhalation, are clearly identifiable and are matching the expected amount of breaths taken.

2.3 Moving Average Filter & Measurement Analysis

The plots in Figure 9 are based on the same raw data then the ones in Figure 8. In this case however, a Moving Average Filter with a window size of 100 samples, has been applied. Using the peaks of the measured Resistance as an identifier for

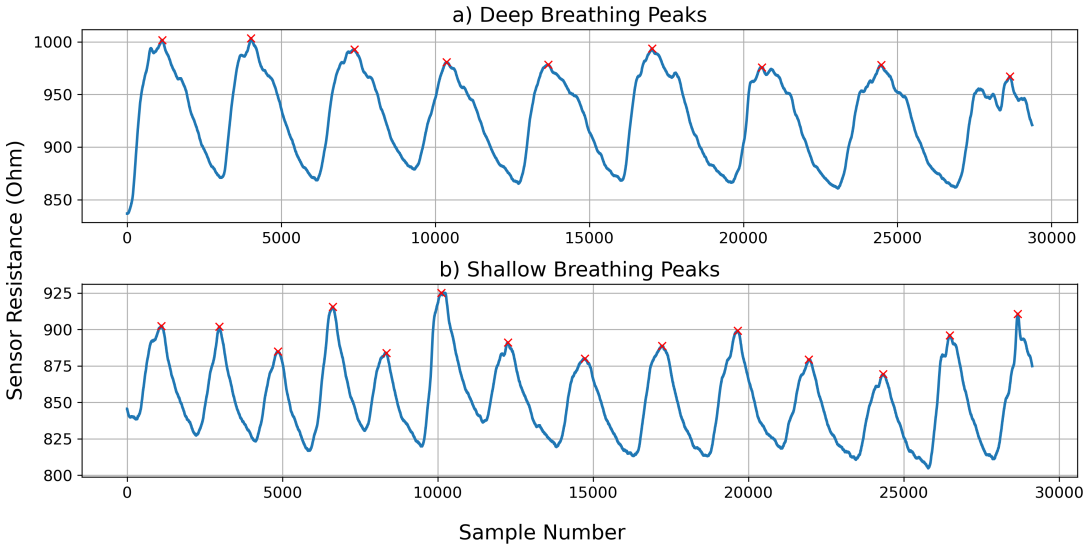


Figure 9: Filtered data of the wearable sensor for varying breathing depth

the number of breaths taken, one could estimate therefore that 9 breaths and 14 breaths were respectively taken during Deep Breathing and Shallow Breathing over the span of about 2 minutes. Such estimation matches the expected number counted while doing the breathing test.